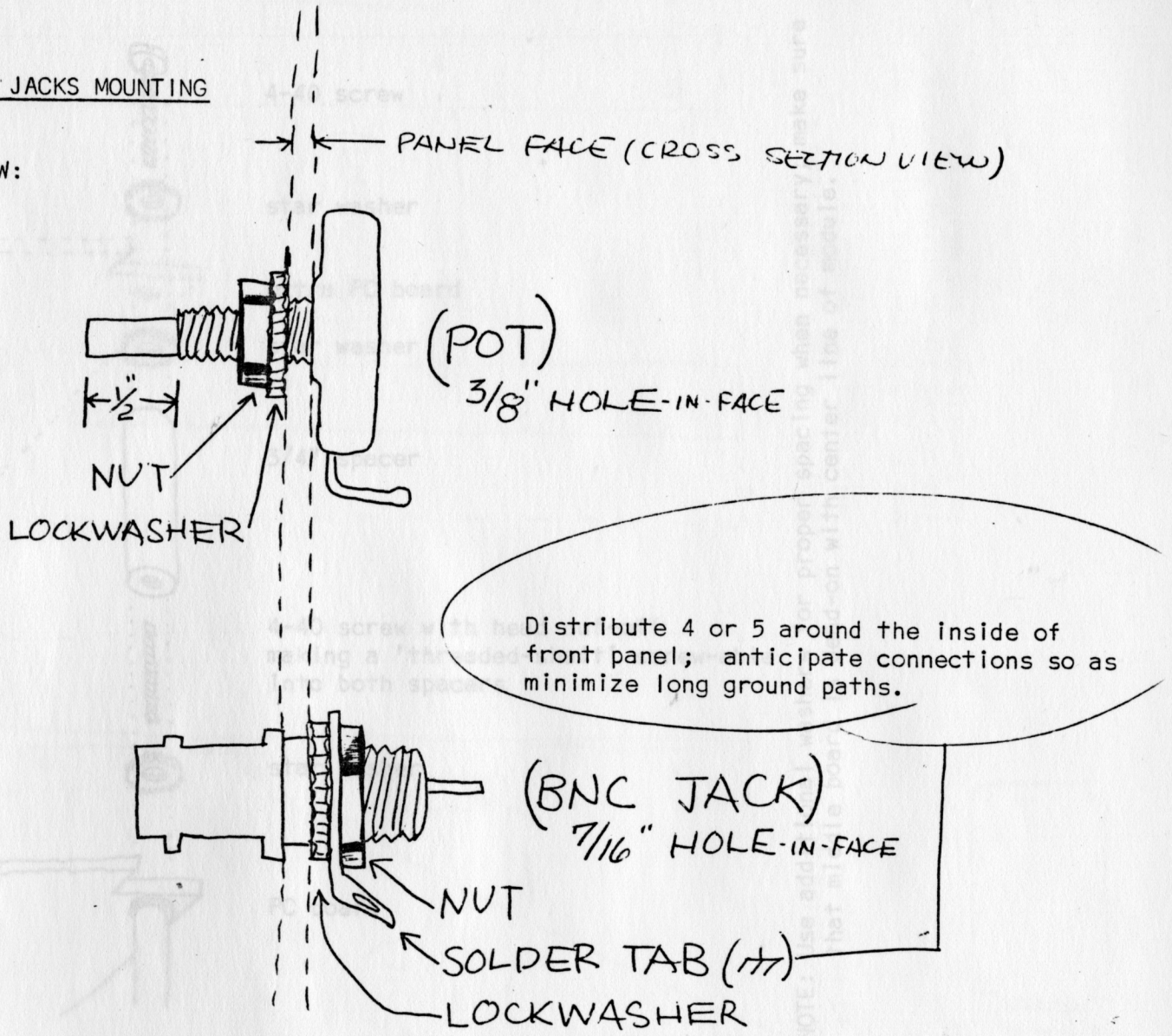
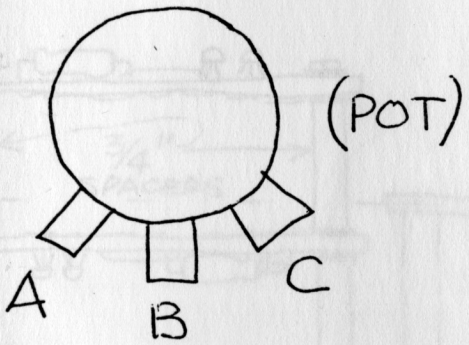
POTS and JACKS MOUNTING

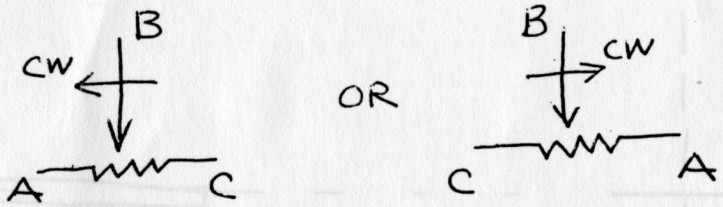
SIDE VIEW:



BACK VIEW:

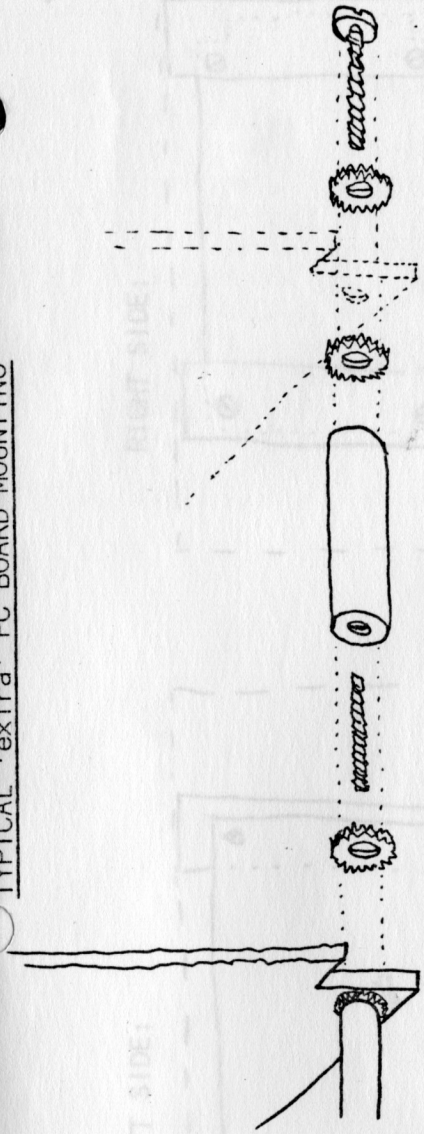


SYMBOL:



CW = clockwise

TYPICAL 'extra' PC BOARD MOUNTING



4-40 screw

star washer

extra PC board

star washer

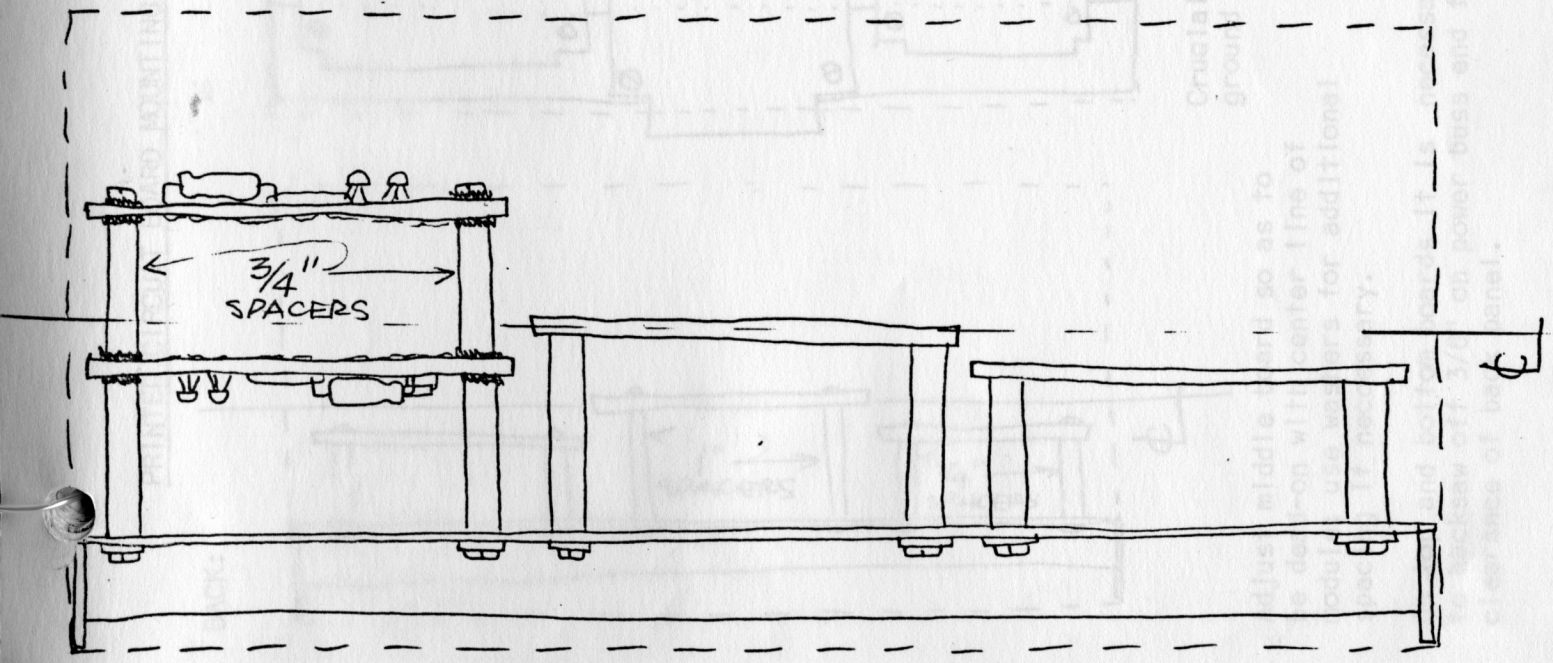
3/4" spacer

4-40 screw with head cut-off
making a 'threaded-shaft' screw-able
into both spacers

star washer

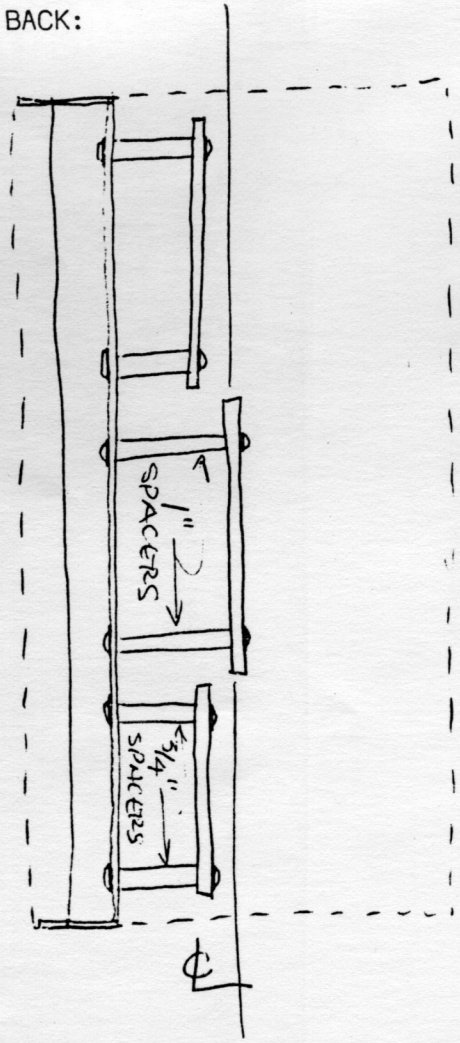
PC board

NOTE: Use additional washers for proper spacing when necessary; make sure that middle board is dead-on with center line of module.



PRINTED CIRCUIT BOARD MOUNTING

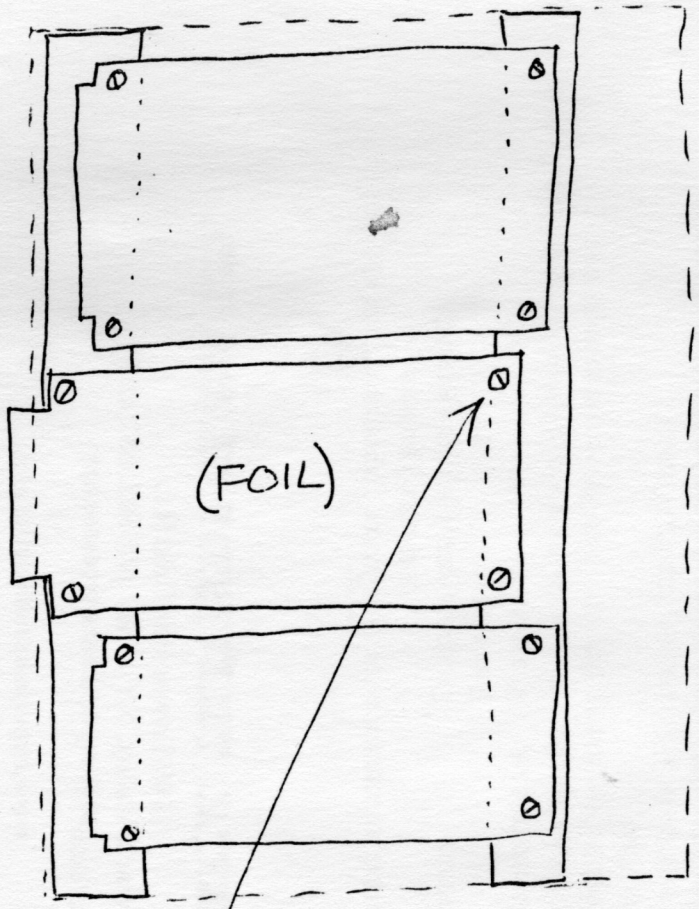
BACK:



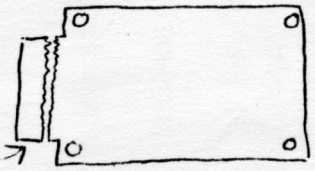
Adjust middle board so as to be dead-on with center line of module; use washers for additional spacing if neccessary.

On top and bottom boards it is necessary to hacksaw off 3/8" on power buss end for clearance of back panel.

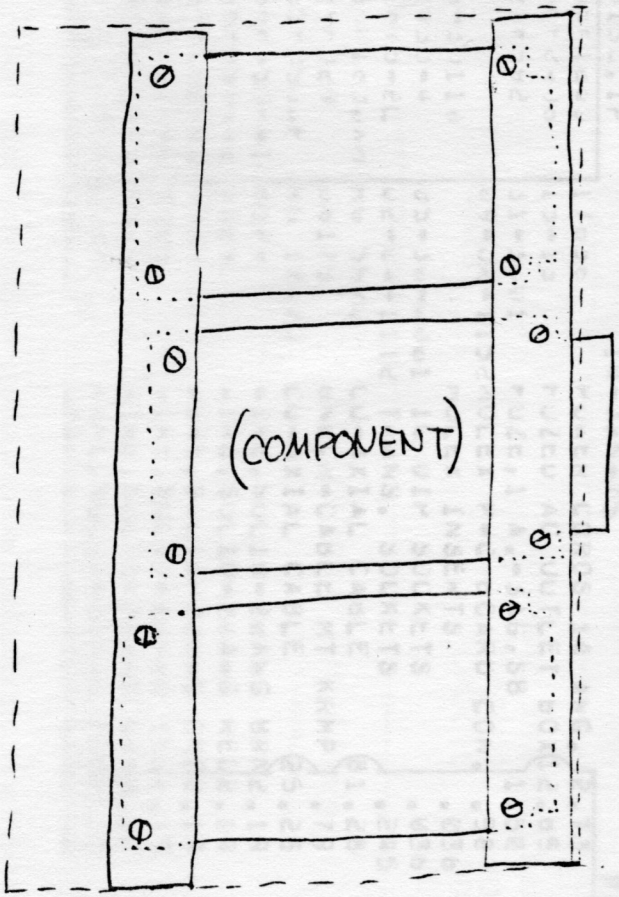
LEFT SIDE:



Crucial screw; circuits all run to ground (⏚) at this point.



RIGHT SIDE:



Star washer all 4-40 screws which hold boards to supports.

Use #6 x 3/8" panhead self thread - ing screws for mounting supports to chassis box (no washers).

ADDER MULTIPLIER:

The adder multiplier is used to add (superimpose), fade and gain control (multiply) signals.

JI1, JI2, JI3 and the inverted signal of JI7 are added together to form input channel A.

JI4, JI5, JI6 and the inverted signal of JI8 are added together to form input channel B.

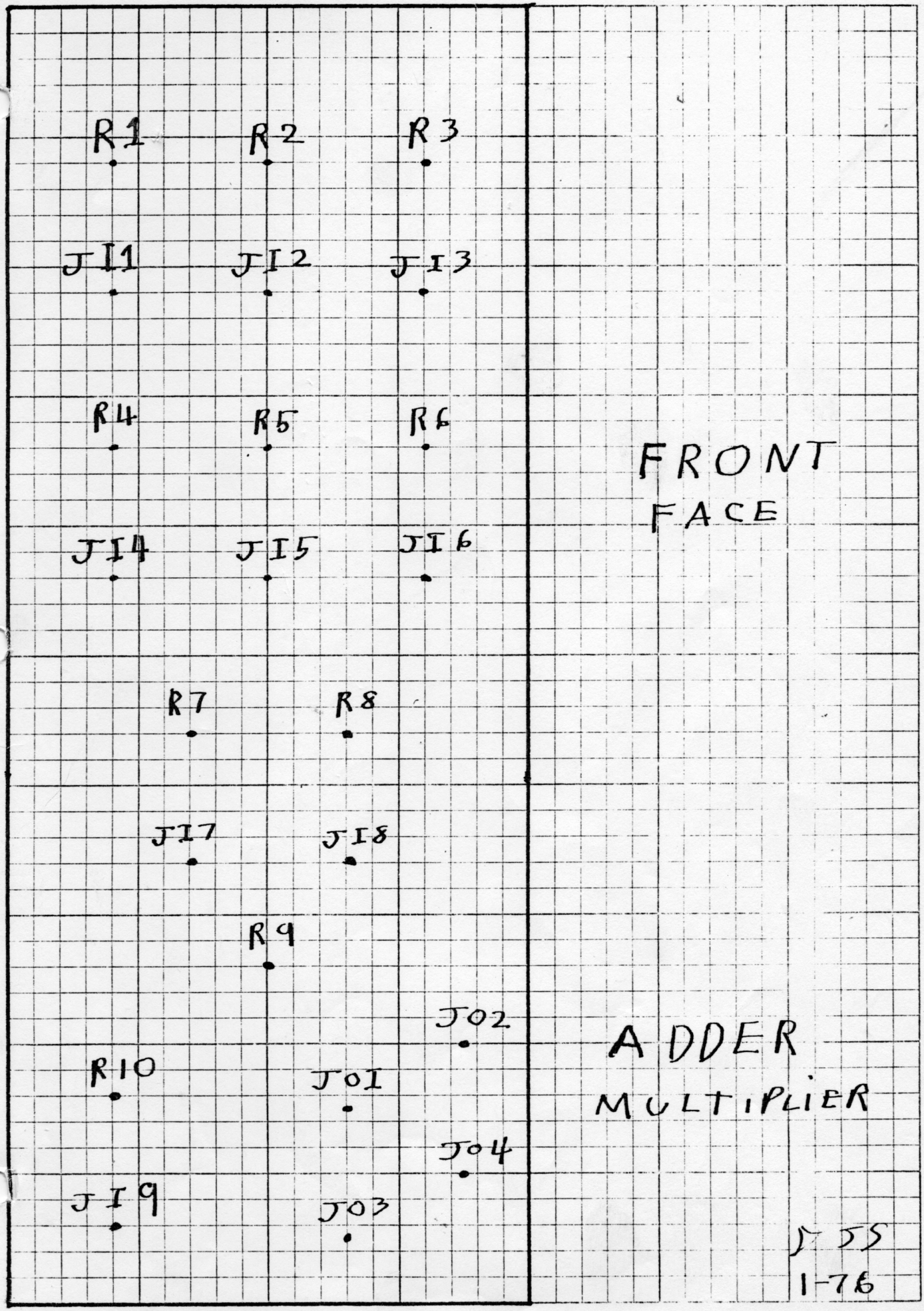
The knobs above the connectors control the gain (contrast) of each individual input.

The amount of channel A and B mixed into the output, JO1 through JO4, is dependent on the position of R9 and the voltage inputted to JI9.
The effect of the knob position and the voltage are additive; the knob to the left and/or a maximum negative voltage on JI9 will cause channel B to be outputted only, similiarly, the knob to the right and/or a maximum positive voltage will cause channel A to be outputted only.
The knob at approximately the center with no voltage applied to JI9 will cause half-of channel A and half-of channel B to be added together and outputted.

TEST STUFF:

The adder multiplier should have a net gain of slightly greater than 1.
That is, a (+) or (-) .5 volt signal into the module should result in an undistorted output of approximately the same magnitude into a 75 ohm load.

With no input the output should be approximately 0 volts (+ or - .05 volts).
Adjust 20k trimmer pot so with R9 in center position and no input to JI9 channel A and channel B have equal gain.


R1
R2
R3
JI1
JI2
JI3
R4
R5
R6
JI4
JI5
JI6
R7
R8
JI7
JI8
R9
JO2
R10
JO1
JO4
JI9
JO3
FRONT
FACE
ADDER
MULTIPLIER
J 55
1-76

MINI ADDER MULTIPLIER:

The mini adder multiplier is used to add (superimpose), fade and gain control (multiply) signals.

Knob center - no output
Knob right of center - non inverted output
Knob left of center - inverted output

J11 is the input to channel A.
J12 is the input to channel B.

The knobs above the connectors control the gain (contrast) of each individual input.

When the knob is turned right of center (12:00 0, Clock) signal increases its' non-inverted gain.

When the knob is turned left of center (12:00 0,Clock) signal increases its inverted gain.

The amount of channel A and B mixed into the output, J01 through J08, is dependent on the position of R3 and the voltage inputted to J13. The effect of the knob position and the voltage are additive; the knob to the left and/or a maximum negative voltage on J13 will cause channel B to be outputted only, similiarly, the knob to the right and/or a maximum positive voltage will cause channel A to be outputted only. The knob at approximately the center with no voltage applied to J13 will cause half-of channel A and half-of channel B to be added together and outputted.

TEST STUFF:

The adder multiplier should have a net gain of slightly greater than 1. That is, a (+) or (-) .5 volt signal into the module should result in and undistorted output of approximately the same magnitude into a 75 ohm load.

With no input the output should be approximately 0 volts (+ or - .05 volts).
Adjust 20k trimmer pot so with R9 in center position and no input to J13 channel A and channel B have equal gain.

'MINI' ADDER/MULTIPLIER:

The 'MINI' ADDER/MULTIPLIER is a packed electrical module containing essentially three of the standard ADDER/MULTIPLIERS (as per documentation). The modification involves the elimination of extra inputs, two each for A and B inputs, and the elimination of the separate inverting inputs as was available on the A/M. The 'MINI' ADDER/MULTIPLIER uses 'bi-polar' inputs, one each for A and B inputs to the 1445L's, enabling that signal input to be 'normal' with increasing gain as R1 or R2 is turned clockwise from center, or 'inverting' with increasing gain as R1 or R2 is turned counter-clockwise from center. Extra standard drivers are used to add four additional outputs per sub-module.

The 'MINI' ADDER/MULTIPLIER can be used to add(superimpose), fade and gain-control (multiply) signals.

Knob center R1,R2--no output (zero volts, middle gray)
Knob right of center R1,R2--non inverting output
Knob left of center R1, R2--inverted output

Knobs above inputs control the gain over that input.

Turning R1,R2 right of center increases the signal input non-inverting gain.

Turning R1, R2 left of center increases the signal input inverted gain.

JI1 is the signal input to channel A.
JI2 is the signal input to channel B.

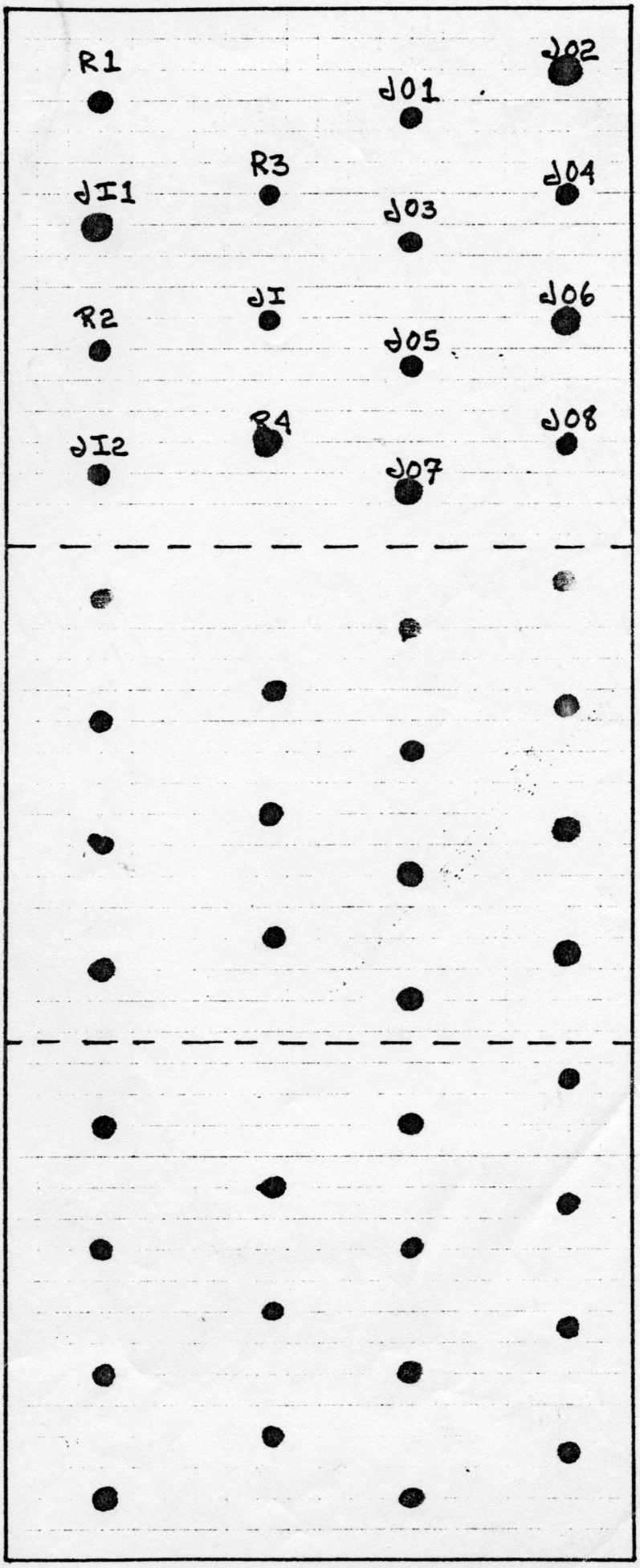
The amount of channel A and B mixed into the output JO1-JO3 is dependent on the position of R4 (BIAS knob) and the voltage inputted to JI3 (multiply input). The effect of the knob position and the voltage at JI3 are additive; the knob to the left and/or a maximum negative voltage at JI3 will cause channel B to be outputted only. Similarily, the knob (R4, Bias) to the right and/or a maximum positive voltage at JI3 will cause channel A to be outputted only. The knob(R4, Bias) at approximately the center with no voltage applied to JI3 will cause half of the signal at A to be added with half of the signal at B and outputted.

TEST STUFF:

The 'MINI' ADDER/MULTIPLIER should have a net gain of slightly greater than 1. That is, a (+) or (-) .5volt signal into the module should result in an undistorted output of approximately the amplitude into 75 ohm load. With no signal input, the output should be zero volts, + or - .05 volts. 'Adjust the 20K trimmer so that with R4 in the center, and precise amplitude signals into A and B (JI1, JI2), and no input to JI3, channel A and B have equal gain at the output.
If you use the 217 board modification, be sure to drill precise holes, cut foil in the correct location, and insulate dc jumper wires.

OK?

11.79 MF



MINI ADDER/ MULTIPLIER